

Baulkham Hills High School

2018

Higher School Certificate
Trial Examination

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- NESA-approved calculators may be used
- A Reference Sheet is provided at the end of this paper.
- In questions 11-16 show relevant mathematical reasoning and/or calculations.

Total Marks 100

Section I 10 marks

- Attempt questions 1 – 10
- Allow about 15 minutes for this section

Section II 90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours and 45 minutes for this section.

Section I

10 marks

Attempt questions 1-10

Allow about 15 minutes for this section.

Use the multiple choice answer sheet for questions 1-10

1. The 40th term of the sequence $-4, 1, 6, \dots$ is:

(A) 25
(B) 61
(C) 155
(D) 191

2. Which of the following expressions is equivalent to $\log_2 7$?

(A) $\frac{\ln 7}{\ln 2}$
(B) $\frac{\ln 2}{\ln 7}$
(C) $2\ln 7$
(D) $7\ln 2$

3. When expressed in terms of π , what is 450° in radians?

(A) $\frac{9\pi}{4}$
(B) 3π
(C) $\frac{7\pi}{5}$
(D) $\frac{5\pi}{2}$

4. Evaluate $\int_0^2 dx$.

(A) 0
(B) 1
(C) 2
(D) None of the above

5. Which of the following could not be the probability of a particular event?

(A) $\frac{2}{3}$

(B) 60%

(C) 0.2378

(D) $\frac{7}{5}$

6. Which of the following statements must be true if a particle is stationary?

(A) The particle's displacement is zero

(B) The particle's velocity is zero

(C) The particle's acceleration is zero

(D) All of the above

7. Sixty tickets are sold in a raffle. There are two prizes. Harry buys five tickets. Which expression gives the probability that Harry wins both prizes?

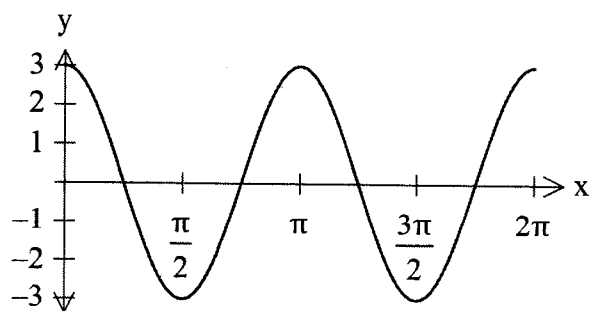
(A) $\frac{5}{60} + \frac{4}{59}$

(B) $\frac{5}{60} + \frac{4}{60}$

(C) $\frac{5}{60} \times \frac{4}{59}$

(D) $\frac{5}{60} \times \frac{4}{60}$

8. The diagram shows a sketch of the graph of $y = f(x)$, for $0 \leq x \leq 2\pi$.



Which of the following equations best describes this curve?

(A) $y = 3\cos 2x$

(B) $y = 2\cos 3x$

(C) $y = 3\cos \frac{x}{2}$

(D) $y = 2\cos \frac{x}{3}$

9. What are the solutions of $\cos 2x = \frac{1}{2}$ for $-\pi \leq x \leq \pi$?

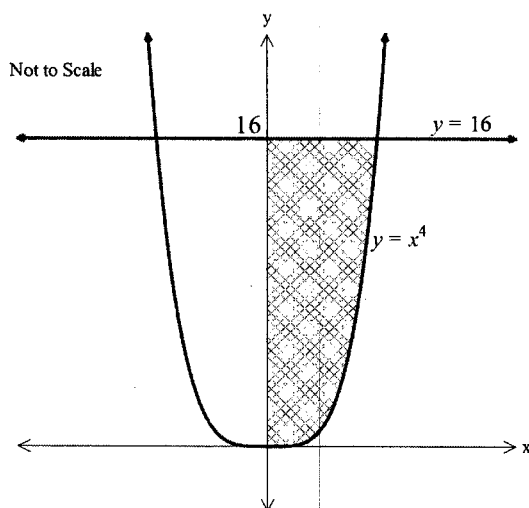
(A) $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{-5\pi}{6}, \frac{-\pi}{6}$

(B) $x = \frac{\pi}{12}, \frac{11\pi}{12}, \frac{-11\pi}{12}, \frac{-\pi}{12}$

(C) $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$

(D) $x = \frac{\pi}{12}, \frac{11\pi}{12}, \frac{13\pi}{12}, \frac{23\pi}{12}$

10. The region in the diagram is bounded by the curve $y = x^4$, the y -axis and the line $y = 16$.



Which of the following expressions is correct for the volume of the solid of revolution when this region is rotated about the y -axis?

(A) $\pi \int_0^2 x^8 dx$

(B) $\pi \int_0^{16} x^8 dx$

(C) $\pi \int_0^2 \sqrt{y} dy$

(D) $\pi \int_0^{16} \sqrt{y} dy$

END OF SECTION I

Section II

90 marks

Attempt questions 11-16

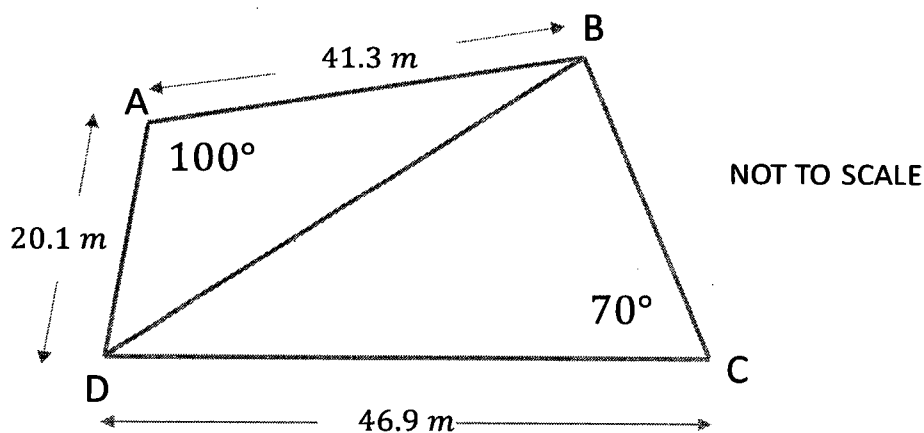
Allow about 2 hours and 45 minutes for this section

Answer each question in the **appropriate page of the writing booklet**. Extra sheets of writing paper are available.

In Questions 11-16, your responses should include relevant mathematical reasoning and/or calculations.

QUESTION 11 (15 marks) – Start on the appropriate page in your booklet.

- a) Evaluate $\sqrt{\pi^2 + 5}$ to two decimal places. 1
- b) Simplify $\frac{x}{x^2-4} - \frac{2}{x-2}$. 2
- c) Solve $|2x - 3| = 4 - 3x$. 3
- d) Consider the expression $(2\sqrt{3} - 1)(3\sqrt{3} + 2) = p + q\sqrt{3}$. Given that p and q are rational numbers, find the values of p and q . 2
- e) Find all the solutions to $4^x - 17 \times 2^x = -16$. 3
- f)



In the above figure, $ABCD$ is a quadrilateral in which $AB = 41.3 \text{ m}$, $AD = 20.1 \text{ m}$, $DC = 46.9 \text{ m}$, $\angle A = 100^\circ$ and $\angle C = 70^\circ$.

- (i) Find the length of BD , giving your answer to two decimal places. 2
- (ii) Find $\angle DBC$, correct to the nearest degree. 2

End of Question 11

QUESTION 12 (15 marks) – Start on the appropriate page in your booklet.

a) Differentiate with respect to x :

(i) $y = x^2 \ln 4x.$ 2

(ii) $y = \frac{\sin 5x}{x^2}.$ 2

b) Find $\int (1 + \sec^2 x) dx$ 1

c) Evaluate the limiting sum of the series

$\frac{3}{4} + \frac{3}{16} + \frac{3}{64} + \dots$ 2

d) Let α and β be the roots of $2x^2 - 4x - 2 = 0$.

(i) State the value of $\alpha\beta$. 1

(ii) Find $\frac{5}{\alpha} + \frac{5}{\beta}$. 2

(iii) Find $\alpha^3 + \beta^3$. 2

e) The number of bacteria in a culture is given by $N = Ae^{kt}$. If 6000 bacteria increase to 9000 after 8 hours,

(i) find k , correct to three significant figures. 2

(ii) find the number of bacteria after 2 days. 1

End of Question 12

QUESTION 13 (15 marks) – Start on the appropriate page in your booklet.

a) Given the equation of the parabola $y = x^2 + 6x + 6$

(i) Find the co-ordinates of the vertex.

1

(ii) Find the co-ordinates of the focus.

1

b) If $f'(x) = 6x^2 + 5x - 1$ and $f(-1) = 5$, find an expression for $f(x)$.

2

c) Consider the function $f(x) = |x - 4|$.

(i) Sketch the function.

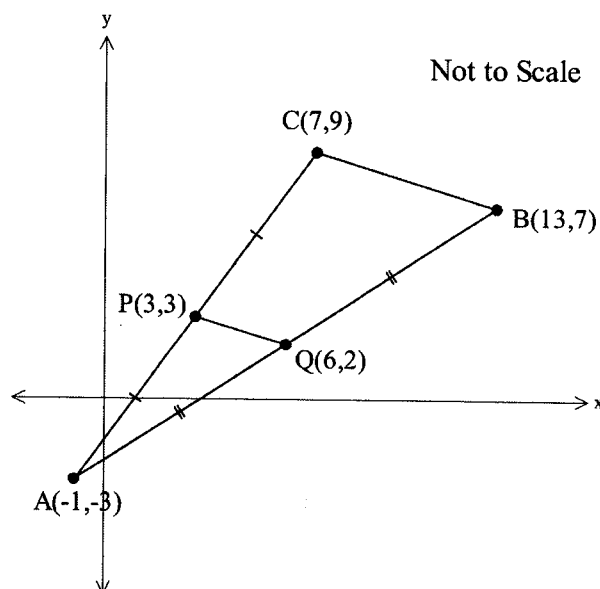
1

(ii) Hence, or otherwise, evaluate $\int_0^6 |x - 4| dx$

2

d) In the diagram A , B and C are the points $(-1, -3)$, $(13, 7)$ and $(7, 9)$ respectively.

The points $P(3, 3)$ and $Q(6, 2)$ are the mid points of AC and AB respectively.



(i) Show that the equation of the line PQ is $x + 3y - 12 = 0$.

2

(ii) Prove that $\triangle ABC$ is similar to $\triangle AQP$.

3

(iii) Find the perpendicular distance of point A to the line PQ .

2

(iv) Hence find the area of the $\triangle AQP$.

1

End of Question 13

QUESTION 14 (15 marks) – Start on the appropriate page in your booklet.

- a) Find the value of m for which the equation $(m - 1)x^2 + 3x - 3 = 0$ has one root twice the other. 3

- b) The line $y = mx + b$ is a tangent to the curve $y = x^2 + 4x + 2$ at $x = -3$.

(i) Find the value of m . 1

(ii) Find the angle of inclination that the tangent makes with the positive x-axis. 2

- c) A sum of \$5000 is deposited at the start of each year in an account that earns 8% per annum, compounded annually.

(i) Show that the amount accrued by the end of the third year can be expressed as

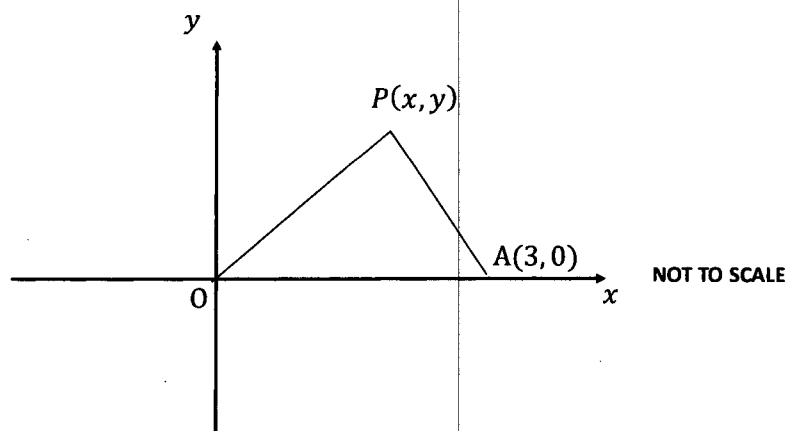
$$A_3 = 5000(1.08 + 1.08^2 + 1.08^3) \quad \text{1}$$

(ii) Show that the amount accrued after n years is given by

$$A_n = 67\,500(1.08^n - 1) \quad \text{1}$$

(iii) Find the total value of the investment at the end of the 15th year, correct to the nearest dollar. 1

- d) Consider the diagram below.

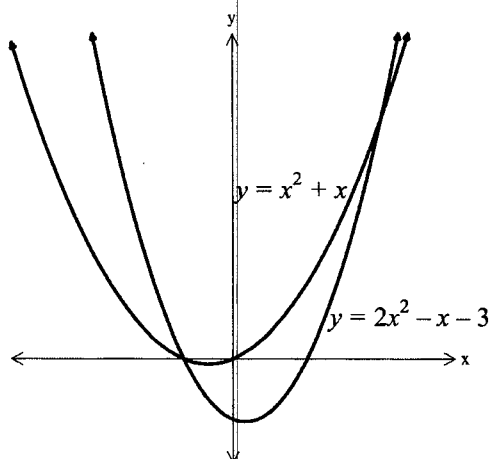


Show that the equation of the locus of all points P , such that $OP \perp AP$ is $x^2 - 3x + y^2 = 0$. 2

Question 14 continues on the following page

QUESTION 14 (continued)

e) Consider the two parabolas as shown below:



- (i) Show that the x values at the points of intersection of $y = x^2 + x$ and $y = 2x^2 - x - 3$, are $x = -1$ and $x = 3$. 1
- (ii) Find the area enclosed between the two curves. 3

End of Question 14

QUESTION 15 (15 marks) – Start on the appropriate page in your booklet.

- a) If $6x^2 - 11 \equiv A(x + 2)^2 + Bx + C$ find the values of A, B and C . 2
- b) Using Simpson's rule, approximate the answer to $\int_1^5 \frac{dx}{x^2 + 1}$, using four sub-intervals. 3
- c) Water was poured into a tank for 30 hours until it was full. At any time, t hours, the volume, V litres, of water in the tank was given by $V = 2(20t - t^2 + 100)$
- (i) How much water was in the tank initially? 1
- (ii) At what rate was water poured into the tank at the end of 5 hours? 2
- d) Consider the function $f(x) = xe^{-x}$.
- (i) Show that $f'(x) = e^{-x}(1 - x)$. 1
- (ii) Show that $f''(x) = e^{-x}(x - 2)$. 1
- (iii) Find the coordinates of any stationary points and determine their nature. 2
- (iv) Find the coordinates of the point of inflexion. 1
- (v) By considering the values of the function as x approaches infinity, sketch $y = f(x)$, indicating all important features found in part (iii) and (iv). 2

End of Question 15

QUESTION 16 (15 marks) – Start on the appropriate page in your booklet.

a) Show that

i) $\frac{4}{x+2} - \frac{2}{x-3} = \frac{2x-16}{x^2-x-6}$ 1

ii) Hence find $\int \frac{2x-16}{x^2-x-6} dx$ 2

b) An airline company marks the price of a flight at \$400, less a group discount based on the number of bookings made. The price R dollars for each person in the group of x people is $R = 400 - 0.5x$. The cost of running the flight is a fixed cost of \$5000 plus \$150 per person.

(i) Show that the profit on a flight of a group of x people is $(250x - 0.5x^2 - 5000)$ dollars. 2

(ii) Hence find the required group size to gain the maximum profit and find this profit. 3

c) Two particles A and B are moving along the x-axis. Their displacements from the origin are given by: $x = -\frac{1}{\pi}(1 + \cos \pi t)$ and $x = t^2 - 4t$ respectively.

i) Express the velocities of the two particles in terms of time. 2

ii) On the same diagram sketch the two velocities. 2

iii) Use that sketch to show that the particles have the same non-zero velocities at two occasions by marking t_1 and t_2 . 1

iv) Show that the distance travelled by the second particle between t_1 and t_2 is: $(t_1^2 + t_2^2) - 4(t_1 + t_2) + 8$ 2

End of Examination

Solutions - Trial - 2 unit 2018

Section one

1. D $T_{40} = 5 \times 40 - 9$

2. A

3. D $450 = 360^\circ + 90^\circ$
radius = $2\pi + \frac{\pi}{2}$
 $= \frac{5\pi}{2}$

4. C $\int_0^1 dx = [x]_0^1$
 $= 2$

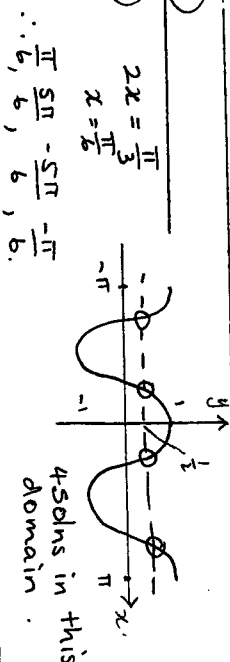
5. D $0 \leq P(E) \leq 1$

6. B

7. C

8. A

9. A $2x = \frac{\pi}{3}$
 $x = \frac{\pi}{6}$



10. D $V = \pi \int_a^b x^2 dy$
 $= \pi \int_0^{16} \sqrt{y} dy$

1 mark each.

Section two

Question 11. (15 marks)

a) $3 \cdot 856 = 3 \cdot 86 (2dp)$

b) $\frac{x}{x^2-4} - \frac{2}{x-2} = \frac{x-2(x+2)}{(x-2)(x+2)}$
 $= \frac{-x-4}{(x-2)(x+2)}$

c) $|2x-3| = 4-3x$

$2x-3 = 4-3x$ $2x-3 = -4+3x$

$x = \frac{7}{5}$ $x = 1$

test: $|2x \frac{7}{5} - 3| = 4 - 3 \times \frac{7}{5}$ $|2 \times 1 - 3| = 4 - 3 \times 1$
 $|- \frac{1}{5}| \neq - \frac{1}{5}$ $|-1| = 1$
no soln. $\therefore x=1$ is the soln.

d) $(2\sqrt{3}-1)(3\sqrt{3}+2) = p+q\sqrt{3}$
 $\therefore 6 \times 3 + 4\sqrt{3} - 3\sqrt{3} - 2$
 $= 16 + \sqrt{3}$ $\therefore p=16$ and $q=1$

e) $4^x - 17 \times 2^x + 16 = 0$
let $m = 2^x$

$\therefore m^2 - 17m + 16 = 0$
 $(m-16)(m-1) = 0$

$\therefore 2^x = 16$ $2^x = 1$
 $= 2^4$ $= 2^0$
 $\therefore x=4$ $x=0$
soln is $x=0$ or 4

Mark

comment

1

correct solution.

2

2 - correct solution
1 - established correct first line

3

3 - correctly solves and tests
 $\therefore x=1$
2 - correctly solves
- no test.
 $\therefore x = \frac{7}{5}, 1$
1 - evidence of two solns.

2

3 - correctly finds both values.
1 - finds either value.

3

3 - a correct solution (any method)
2 - attempts to solve with 2 solns.
1 - one correct solution.
1 - incorrect substitution completing to 2 solns.

Quest. 11. continued

f. i) $BD^2 = AD^2 + AB^2 - 2 \times AD \times AB \times \cos B$
 $= 20.1^2 + 41.3^2 - 2 \times 20.1 \times 41.3 \times \cos 100$
 $= 2398.0011 \dots$
 $BD = 48.9693 \dots$
 $= 48.97 \text{ (2dp) m.}$

ii) $\frac{\sin \hat{D}\hat{B}\hat{C}}{DC} = \frac{\sin C}{BD}$
 $\sin \hat{D}\hat{B}\hat{C} = \frac{46.9 \times \sin 70}{48.969 \dots}$
 $= 0.899 \dots$
 $\hat{D}\hat{B}\hat{C} = 64.155 \dots$
 $\hat{D}\hat{B}\hat{C} \hat{=} 64^\circ$

Question 12 (15 marks)

a) $\frac{dy}{dx} = 2x \ln 4x + 2 \times \frac{4}{4x}$
 $= 2x \ln 4x + 2$

ii) $y = \frac{\sin 5x}{x^2}$
 $\frac{dy}{dx} = \frac{x^2 \times 5 \cos 5x - \sin 5x \times 2x}{x^4}$
 $= \frac{5x \cos 5x - 2 \sin 5x}{x^3}$

b) $\int 1 + \sec^2 x \, dx = x + \tan x + c$

c) $a = \frac{3}{4}$ $r = \frac{1}{4}$
 $\lim_{S \rightarrow 1} S = \frac{a}{1-r}$
 $= \frac{\frac{3}{4}}{1 - \frac{1}{4}}$
 $= 1$

quest. 12. cont.

d) $2x^2 - 4x - 2 = 0$
 i) $x \propto \beta = \frac{c}{a}$
 $= \frac{-2}{2} = -1$

ii) $\frac{5}{\alpha} + \frac{5}{\beta} = \frac{5(\alpha + \beta)}{\alpha\beta}$
 $= \frac{5 \times 2}{-1}$
 $= -10$
 $\alpha + \beta = \frac{-b}{a}$
 $= \frac{4}{2} = 2$

iii) $\alpha^3 + \beta^3 = (\alpha + \beta)(\alpha^2 - \alpha\beta + \beta^2)$
 $= (\alpha + \beta)((\alpha + \beta)^2 - 3\alpha\beta)$
 $= 2(2^2 - 3 \times -1)$
 $= 14$

e) $N = Ae^{kt}$
 $t=0 \quad N=6000 = Ae^0$
 $\therefore A = 6000$
 $N = 6000e^{kt}$
 $t=8$
 $9000 = 6000e^{\frac{8k}{8}}$
 $\frac{3}{2} = e$
 $\ln \frac{3}{2} = 8k \ln e$
 $k = \frac{\ln 1.5}{8}$
 $= 0.05068 \dots$
 $= 0.0507 \text{ (3 sig fig)}$

ii) 2 days = 48 hrs
 $N = 6000e^{\frac{48k}{8}}$
 $= 68343.75$
 $\hat{=} 68343 \text{ bacteria}$

mark	comment
2.	② - correct answer - ignore dec. places. ① sets up the cosine rule correctly.
2.	② correct soln to nearest degree ① sets up the sine rule correctly.
2.	② correct soln. ① attempts product rule showing $\frac{4}{x^2} \times \frac{4}{4x}$ ③ correct soln ① attempts quotient rule.
1	① correct soln 'ignore +c'
2	② finding lims correctly ① identifying $r = \frac{1}{4}$.

mark	comment
1	② correct value
2	② correct value ① evidence of correct method. ① finds $\alpha + \beta = 2$.
2	② correct value ① evidence of factorising cubic
2	② correct k to 3 sig fig ① evidence evaluating k from correct A.
1	① correct answer - approx. ok.

Question 13 (15 marks)

mark	comment
1	① correct vertex
1	① correct focus
2	② correctly find $f(x)$ ① forgets 'c'
1	must use ruler! and show $y=4$, $x=4$
2	③ correct Area = 10 u ² ① evidence of splitting the area up and correctly finds one Area
2	③ evidence of method. - can sub points in but must show both work. ① finds gradient

a) $y = x^2 + 6x + 6$
 ii) $x^2 + 6x + 9 = y + 9 - 6$
 $(x+3)^2 = y+3$

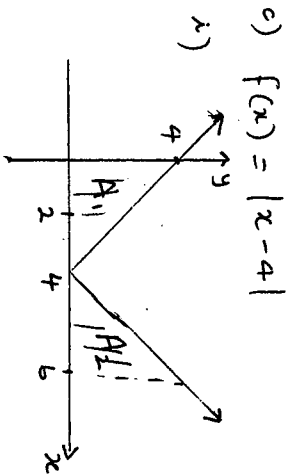
Vertex: $(-3, -3)$

ii) focus: $(-3, -2\frac{3}{4})$ so $4a = 1$
 $a = \frac{1}{4}$

b) $f'(x) = 6x^2 + 5x - 1$
 $f(x) = 2x^3 + \frac{5}{2}x^2 - x + c$

now $f(-1) = -2 + \frac{5}{2} + 1 + c = 5$
 $\therefore c = \frac{7}{2}$

$f(x) = 2x^3 + \frac{5}{2}x^2 - x + \frac{7}{2}$



ii) $\int_0^6 |x-4| dx = A_1 + A_2$
 $= \frac{1}{2} \times 4 \times 4 + \frac{1}{2} \times 2 \times 2$
 $= 10 \text{ u}^2$

d) i) $M_{pq} = \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{2-3}{6-3}$

$= -\frac{1}{3}$

$y - y_1 = m(x - x_1)$ where (x, y) is $(6, 2)$
 $y - 2 = -\frac{1}{3}(x - 6)$

$x + 3y - 12 = 0$

quest 13 cont.

d) ii) Aim: to prove $\triangle ABC \parallel \triangle AQP$
 Proof: In $\triangle ABC$ and $\triangle AQP$

$\hat{A}$ is common

$\frac{AP}{AB} = \frac{1}{2}$ (P is the mid pt - given)

$\frac{AQ}{AC} = \frac{1}{2}$ (Q is mid pt - given)

$\therefore \triangle ABC \parallel \triangle AQP$ (two sides in ratio included $\angle$ equal)

iii) perp d = $\frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$ $A(-1, -3)$
 line: $x + 3y - 12 = 0$

$= \frac{|1 \times -1 + 3 \times -3 - 12|}{\sqrt{1+9}}$

$= \frac{|-22|}{\sqrt{10}}$

$= \frac{22}{\sqrt{10}}$ or $\frac{11\sqrt{10}}{5}$ units.

iv. $d_{PQ} = \sqrt{(3-6)^2 + (3-2)^2}$

$= \sqrt{10}$

$A = \frac{1}{2} \times \frac{22}{\sqrt{10}} \times \sqrt{10}$

$= 11 \text{ u}^2$

mark comment

③ clearly structured proof. "included" stated

② - final reason poorly stated but body supports proof. - ratio not clearly defined
 ① made an attempt
 $\hat{A}$ is common

③ correct answer
 ① correct substitution

correct result.
 (no problem if units missing)

Question 14 (15 marks).

a) $(m-1)x^2 + 3x - 3 = 0$
 let the roots be α and 2α
 $\therefore$ sum: $\alpha + 2\alpha = -\frac{b}{a}$

$$3\alpha = -\frac{3}{m-1} \dots \textcircled{1}$$

product: $\alpha \times 2\alpha = \frac{c}{a}$
 $2\alpha^2 = \frac{-3}{m-1} \dots \textcircled{2}$

rearrange $\textcircled{1}$
 $\alpha = \frac{-1}{m-1}$

sub into $\textcircled{2}$
 $2\left(\frac{-1}{m-1}\right)^2 = \frac{-3}{m-1}$

$$2 = -3(m-1)$$

$$m = \frac{1}{3}$$

b) $y = mx + b$ $y = x^2 + 4x + 2$ at $x = -3$

i) $\frac{dy}{dx} = 2x + 4$ $m = 2x - 3 + 4 = -2$

ii) $\tan \theta = m = -2$
 $\theta = 180 - 63.43 = 116.57$
 $\hat{=} 117^\circ$ (nearest degree)



mark comment

3 $\textcircled{3}$ correct answer.
 - may use another method.

$\textcircled{2}$ sum and product clearly shown and a substitution made.

$\textcircled{1}$ sum & product of roots attempted

correct value (any method)

$\textcircled{2}$ correct answer.
 - doesn't need to be nearest degree
 $\textcircled{1}$ states $\tan \theta = m = -2$

quest 14 cont.

c) $\$5000$ $r = 0.08$
 $A = P \times (1+r)$
 i) $A_1 = 5000(1+0.08)$
 End of yr 2.

$A = A_1(1.08) + A_2$
 $= 5000(1.08) + 5000(1.08)$
 $= 5000(1.08 + 1.08^2)$
 end of yr 3
 $A = 5000(1.08 + 1.08^2 + 1.08^3)$

ii) $A_n = 5000(1.08 + 1.08^2 + 1.08^3 + \dots + 1.08^n)$
 G.P.
 $a = 1.08$ $r = 1.08$

$$\therefore A_n = 5000 \times \frac{1.08(1.08^n - 1)}{1.08 - 1}$$

$$= 67500(1.08^n - 1)$$

iii) when $n = 15$

$$A_{15} = 67500(1.08^{15} - 1)$$

$$= \$146621.42$$

$$= \$146621 \text{ (nearest dollar)}$$

d) $M_{op} = \frac{y_2 - y_1}{x_2 - x_1}$
 $= \frac{y - 0}{x - 0}$
 $= \frac{y}{x}$

$$M_{PA} = \frac{y - 0}{x - 3}$$

$$= \frac{y}{x - 3}$$

$$= \frac{1}{1 - \frac{3}{x}}$$

now $OP \perp AP \Rightarrow M_{op} = -M_{PA}$

$$\frac{y}{x} = -\frac{1}{1 - \frac{3}{x}}$$

$$y^2 = -x^2 + 3x$$

$$x^2 - 3x + y^2 = 0$$

mark

1.

comment
 must clearly show development from 1st to 2nd to 3rd yr.

clearly show working towards (ie. 2nd last line!)

accept not rounded

$\textcircled{2}$ clear working showing $m_1 = -\frac{1}{m_2}$
 $\textcircled{1}$ finds gradients.

2

quest 14 cont.

e) i) $y = x^2 + x = 2x^2 - x - 3$

$x^2 - 2x - 3 = 0$

$(x-3)(x+1) = 0$

$x = -1, 3$

comment

- can sub in both points or solve

quest 15 cont.

c) $V = 2(20t - t^2 + 100)$

i) $t = 0$

$\therefore V = 2(0 + 100) = 200 \text{ L}$

mark

comment

correct answer - ignore units

ii) $A = \int_{-1}^3 x^2 + x - (2x^2 - x - 3) dx$

$= \int_{-1}^3 (-x^2 + 2x + 3) dx$

$= \left[-\frac{x^3}{3} + x^2 + 3x \right]_{-1}^3$

$= (-9 + 9 + 9) - \left(\frac{1}{3} + 1 - 3 \right)$

$= \frac{32}{3} \text{ v}^2$

③ complete solution.

② sets up integrals and integrate - makes calculation errors

① wrong way round for integrals but reaches an answer
① clearly 2 errors but method correct.

Question 15 (15 marks)

a) $6x^2 - 11 \equiv A(x+2) + Bx + C$

RHS = $Ax^2 + 4Ax + 4A + Bx + C$

$= Ax^2 + x(4A + B) + 4A + C$

now

$6 = A$ $0 = 4A + B$ $-11 = 4A + C$

$0 = 24 + B$ $-11 = 24 + C$

$B = -24$ $C = -35$

$\therefore A = 6, B = -24, C = -35$

② correct solution

① 2 solutions correct

b) $\int_1^5 \frac{dx}{x^2+1}$

$\frac{x}{1} \mid \frac{1}{2} \mid \frac{2}{5} \mid \frac{3}{10} \mid \frac{4}{17} \mid \frac{5}{26}$

$= \frac{1}{3} \left\{ 1st + last + 4 \text{ even} + 2 \times odd \right\}$

$= \frac{1}{3} \left\{ \frac{1}{2} + \frac{1}{26} + 4 \left(\frac{1}{5} + \frac{1}{17} \right) + 2 \times \frac{1}{10} \right\}$

$= \frac{1}{3} \times \frac{392}{221}$

$= 0.5912 \dots$

③ correct solution

② apply Simpson's rule correctly

① significant progress towards solution

ii) $V = 2(20t - t^2 + 100)$

$\frac{dV}{dt} = 2(20 - 2t)$

at $t = 5$

$\frac{dV}{dt} = 2(20 - 10) = 20 \text{ L/h}$

2

③ correct solution including units

① correct derivative or incorrect derived but sub in $t=5$.

d) $f(x) = xe^{-x}$

i) $f'(x) = e^{-x} - xe^{-x}$

$= e^{-x}(1-x)$

ii) $f''(x) = -e^{-x} + xe^{-x} - e^{-x}$

$= xe^{-x} - 2e^{-x}$

$= e^{-x}(x-2)$

1

Shows product rule clearly.

iii) stationary pts $f'(x) = 0$

$\therefore f'(x) = e^{-x}(1-x) = 0$

since $e^{-x} > 0$ $1-x=0$ $x=1$ $y=e^{-1}$ $(1, e^{-1})$

test table or 2nd derivative.

$f''(x) = e^{-1}(1-1)$

$= -e^{-1} < 0 \therefore$ max at $x=1$

2

② correctly find pt and tests.

① wrong pt and tests

③ states $e^{-x} > 0 \therefore x=$

iv) point of inflexion: $f''(x) = 0$

$f''(x) = e^{-x}(x-2) = 0$

since $e^{-x} > 0 \therefore x-2=0$ $x=2$ $y=2e^{-2}$ $(2, 2e^{-2})$

test: - good practice!

$f''(1.5) = e^{-1.5}(1.5-2)$

$= -0.11 \dots < 0$

$f''(2.5) = e^{-2.5}(2.5-2)$

$= 0.04 \dots > 0$

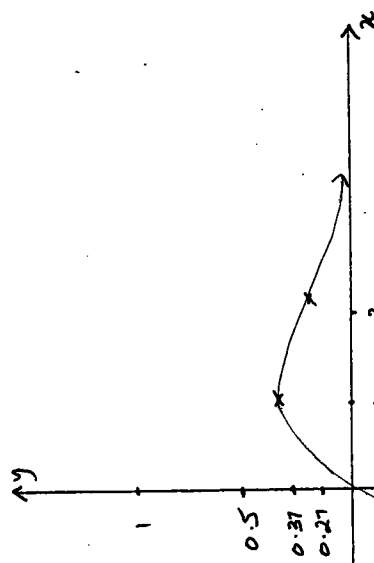
$\therefore (2, 2e^{-2})$ is a point of inflexion

1

finds point giving x and y

quest 15 cont.

V.



$$\begin{aligned} \text{as } x \rightarrow \infty \quad e^{-x} &= \frac{1}{e^x} \rightarrow 0 \quad \therefore f(x) = xe^{-x} \rightarrow 0 \\ \text{at } x = -3 \quad xe^{-(-3)} &= -60 \end{aligned}$$

Question 16 (15 marks)

$$a) \frac{4}{x+2} - \frac{2}{x-3} = \frac{2x-16}{x^2-x-6}$$

$$b) \frac{4}{x+2} = \frac{4}{x+2} - \frac{2}{x-3}$$

$$= \frac{4x-12-2(x+2)}{(x+2)(x-3)}$$

$$= \frac{2x-16}{x^2-x-6} \text{ as required.}$$

$$ii) \int \frac{2x-16}{x^2-x-6} dx = \int \frac{4}{x+2} - \frac{2}{x-3} dx$$

$$= 4 \ln(x+2) - 2 \ln(x-3) + C$$

$$\text{or } = \ln \frac{(x+2)^4}{(x-3)^2} + C$$

Mark

comment

2

③ detail clearly shown - including approaching x axis.

① start pt of inflation - smooth curve shown.

clearly shows

③ correct form of soln.

① correct split for integration ignore +c.

quest 16 cont.

b) Price = \$400/person

i) $R = 400 - 0.5x$ for each person

$$\text{Cost} = \$5000 + \$150x = 5000 + 150x$$

$$\begin{aligned} \text{Profit} &= \text{Price} - \text{cost} \quad \text{Price} = R \times x \\ &= (400 - 0.5x)x - (5000 + 150x) \\ &= 400x - 0.5x^2 - 5000 - 150x \\ &= 250x - 0.5x^2 - 5000 \end{aligned}$$

$$ii) p = 250x - 0.5x^2 - 5000$$

$$\frac{dp}{dx} = 250 - 2 \times 0.5x$$

$$= 250 - x = 0$$

$$x = 250 \text{ people}$$

test:

$$\frac{d^2p}{dx^2} = -1 < 0 \therefore \text{max}$$

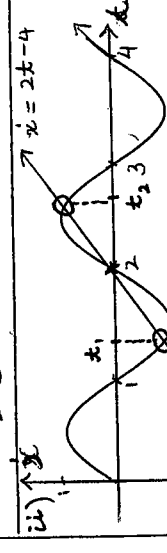
$\therefore$ maximum group size is 250 people.

$$\text{Profit} = 250 \times 250 - 0.5 \times 250^2 - 5000 = \$26250$$

$$c. A: x = -\frac{1}{\pi}(1 + \cos \pi t) \quad B: x = t^2 - 4t$$

$$i) \text{ velocity } \dot{x} = -\frac{1}{\pi}(-\pi \sin \pi t) = \sin \pi t$$

$$\dot{x} = 2t - 4$$



iii) x_1, x_2

Mark

comment

1

x_1, x_2 clearly indicated

quest 1b cont.

$$\begin{aligned}
 \text{iv) Dist} &= \left| \int_{x_1}^2 2x-4 \, dx \right| + \int_2^{x_2} 2x-4 \, dx \\
 &= \left| \left[x^2-4x \right]_{x_1}^2 \right| + \left[x^2-4x \right]_2^{x_2} \\
 &= \left| 4-8-(x_1^2-4x_1) \right| + ((x_2^2-4x_2-4+8)) \\
 &= \left| -(4+x_1^2-4x_1) \right| + x_2^2-4x_2+4 \\
 &= 4+x_1^2-4x_1+x_2^2-4x_2+4 \\
 &= (x_1^2+x_2^2)-4(x_1+x_2)+8 \quad \text{as required}
 \end{aligned}$$

mark	comment
2	② correct soln. ① correctly splits area.